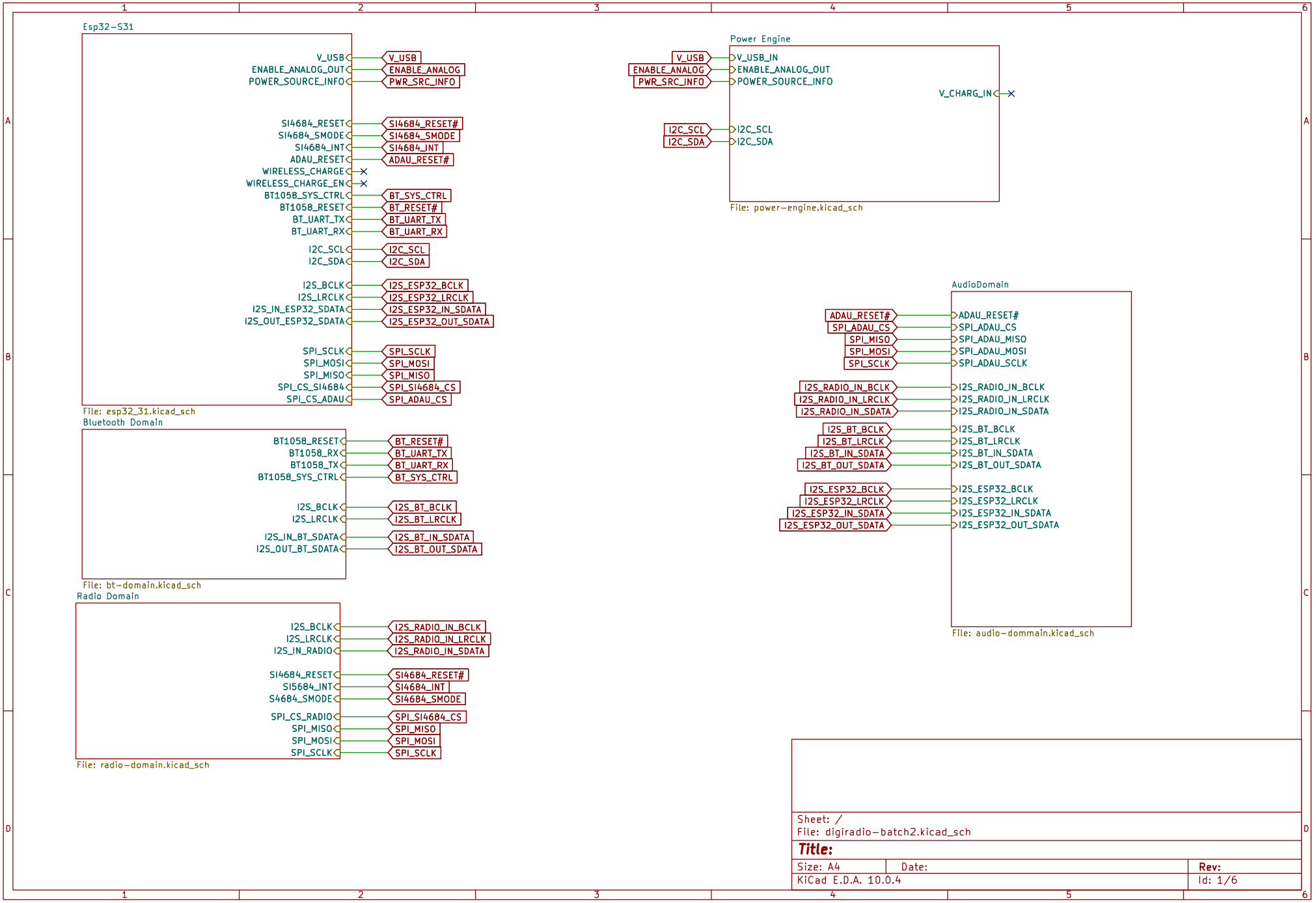
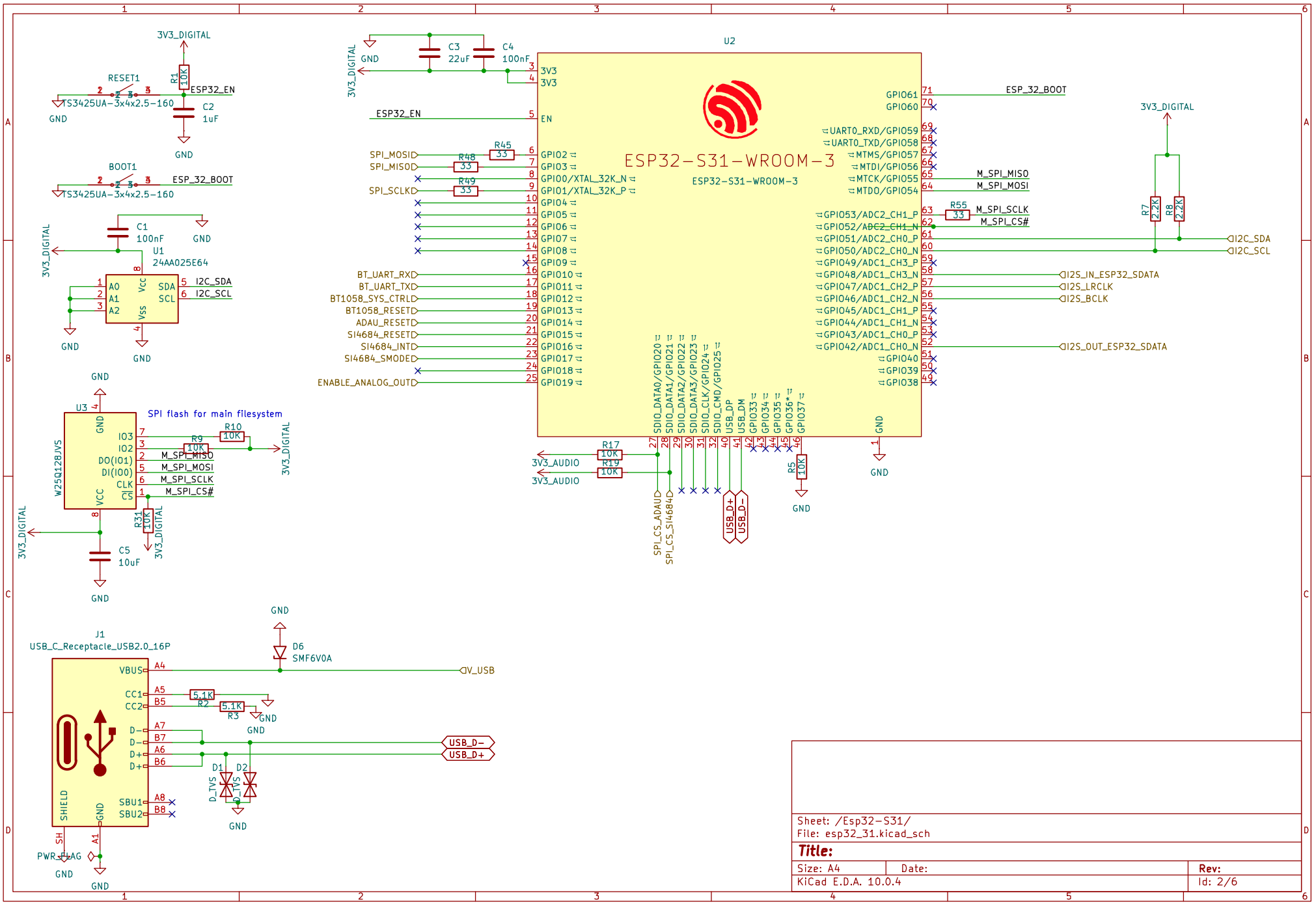
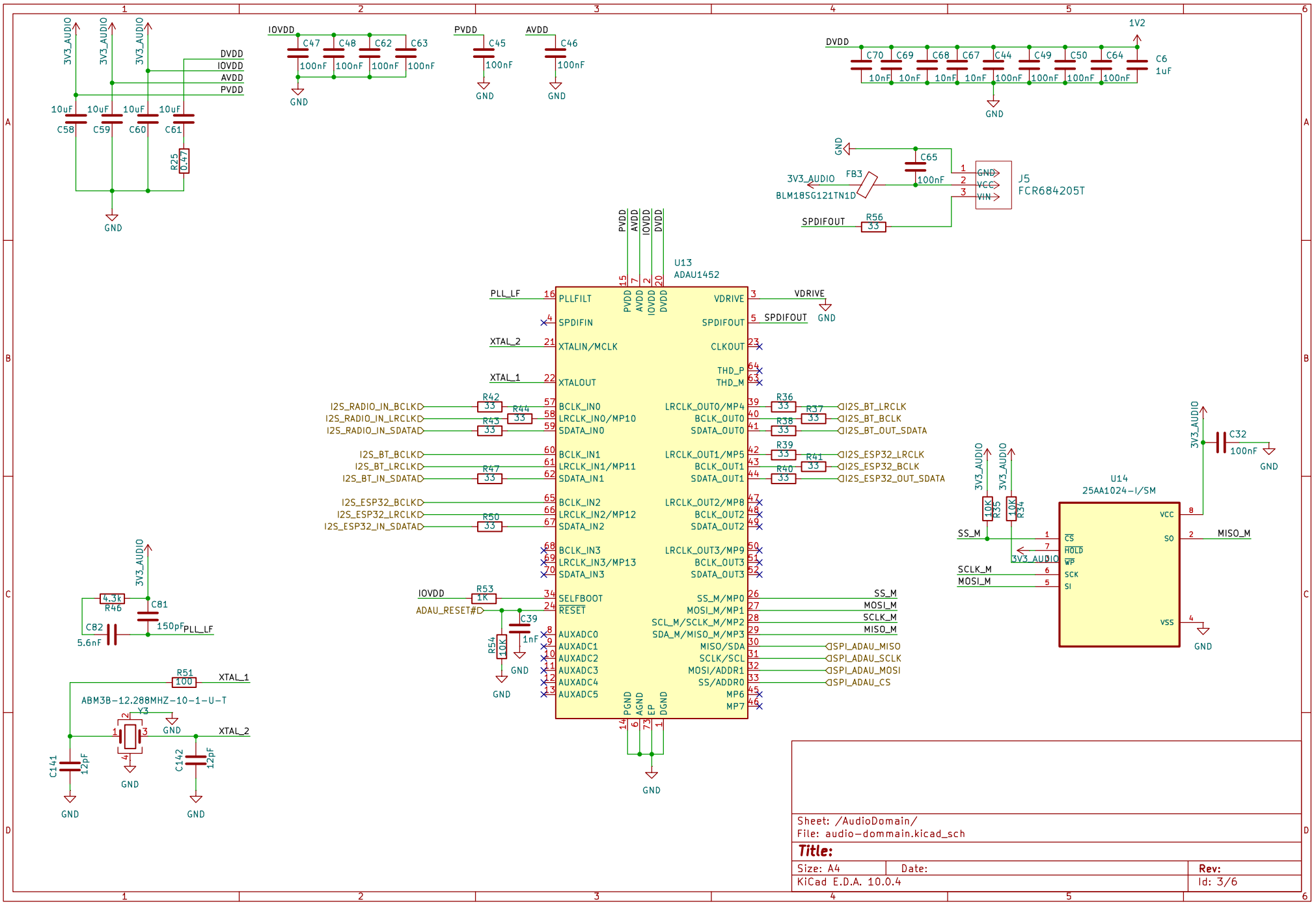
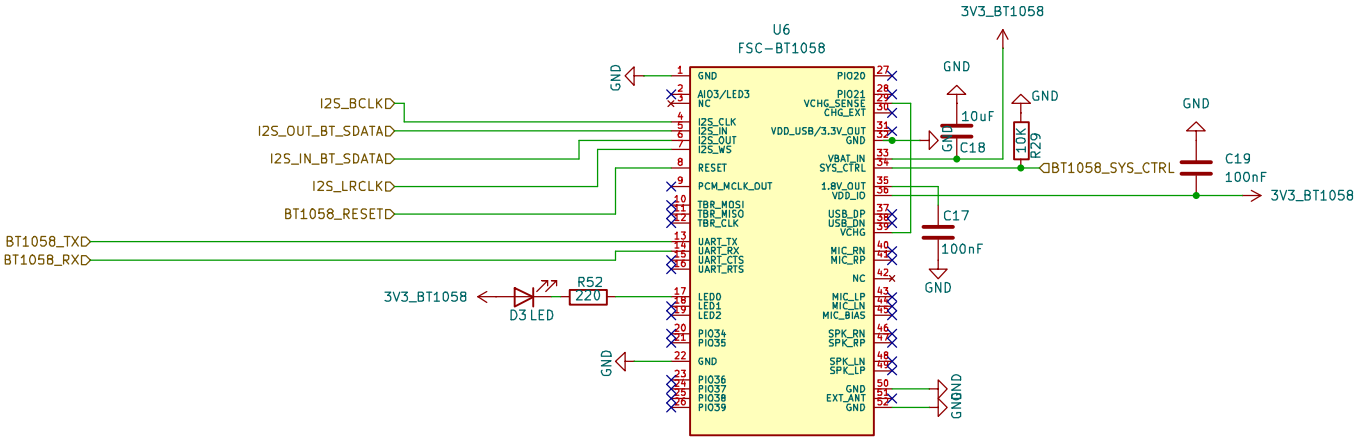
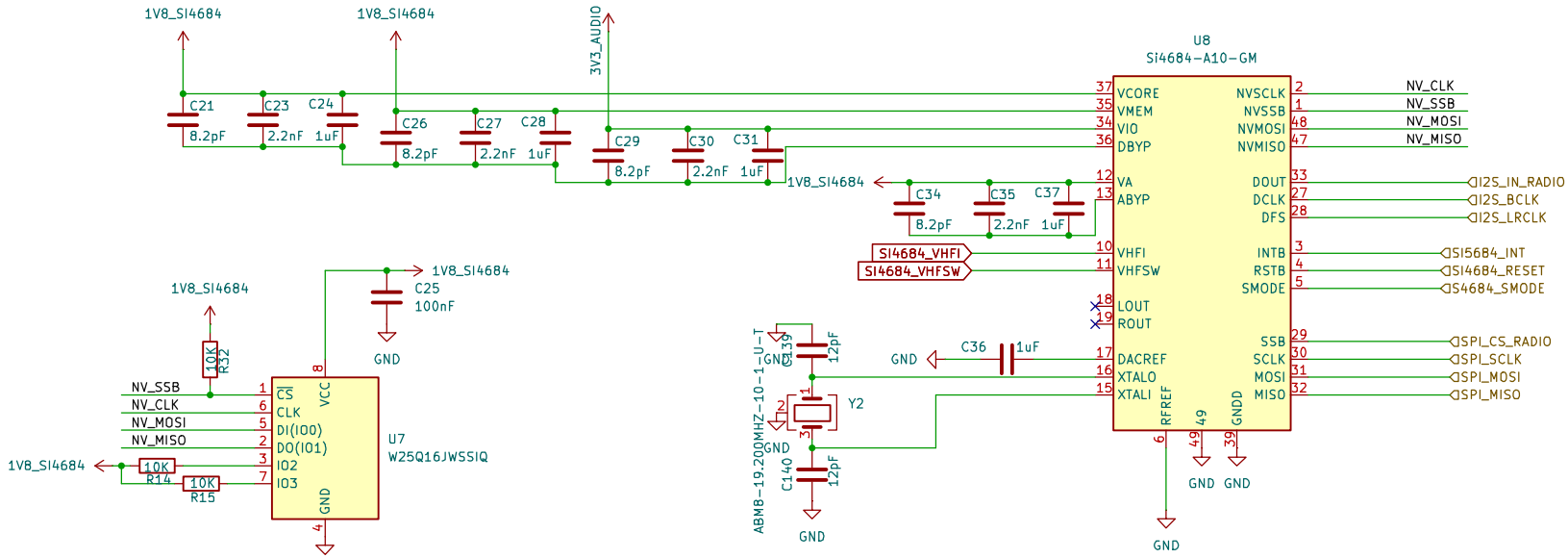

Esp32-S31
V_USB
ENABLE_ANALOG_OUT
POWER_SOURCE_INFO
SI4684_RESET
SI4684_SMODE
SI4684_INT
ADAU_RESET
WIRELESS_CHARGE
WIRELESS_CHARGE_EN
BT1058_SYS_CTRL
BT1058_RESET
BT_UART_TX
BT_UART_RX
I2C_SCL
I2C_SDA
I2S_BCLK
I2S_LRCLK
I2S_IN_ESP32_SDATA
I2S_OUT_ESP32_SDATA
SPI_SCLK
SPI_MOSI
SPI_MISO
SPI_CS_SI4684
SPI_CS_ADAU
V_USB
ENABLE_ANALOG
PWR_SRC_INFO
SI4684_RESET#
SI4684_SMODE
SI4684_INT
ADAU_RESET#
BT_SYS_CTRL
BT_RESET#
BT_UART_TX
BT_UART_RX
I2C_SCL
I2C_SDA
I2S_ESP32_BCLK
I2S_ESP32_LRCLK
I2S_ESP32_IN_SDATA
I2S_ESP32_OUT_SDATA
SPI_SCLK
SPI_MOSI
SPI_MISO
SPI_SI4684_CS
SPI_ADAU_CS
File: esp32_31.kicad_sch
Power Engine
V_USB
ENABLE_ANALOG
PWR_SRC_INFO
I2C_SCL
I2C_SDA
V_USB_IN
ENABLE_ANALOG_OUT
POWER_SOURCE_INFO
V_CHARG_IN
I2C_SCL
I2C_SDA
File: power-engine.kicad_sch
Bluetooth Domain
BT1058_RESET
BT1058_RX
BT1058_TX
BT1058_SYS_CTRL
I2S_BCLK
I2S_LRCLK
I2S_IN_BT_SDATA
I2S_OUT_BT_SDATA
BT_RESET#
BT_UART_TX
BT_UART_RX
BT_SYS_CTRL
I2S_BT_BCLK
I2S_BT_LRCLK
I2S_BT_IN_SDATA
I2S_BT_OUT_SDATA
File: bt-domain.kicad_sch
Radio Domain
I2S_BCLK
I2S_LRCLK
I2S_IN_RADIO
SI4684_RESET
SI5684_INT
S4684_SMODE
SPI_CS_RADIO
SPI_MISO
SPI_MOSI
SPI_SCLK
I2S_RADIO_IN_BCLK
I2S_RADIO_IN_LRCLK
I2S_RADIO_IN_SDATA
SI4684_RESET#
SI4684_INT
SI4684_SMODE
SPI_SI4684_CS
SPI_MISO
SPI_MOSI
SPI_SCLK
File: radio-domain.kicad_sch
AudioDomain
ADAU_RESET#
SPI_ADAU_CS
SPI_MISO
SPI_MOSI
SPI_SCLK
I2S_RADIO_IN_BCLK
I2S_RADIO_IN_LRCLK
I2S_RADIO_IN_SDATA
I2S_BT_BCLK
I2S_BT_LRCLK
I2S_BT_IN_SDATA
I2S_BT_OUT_SDATA
I2S_ESP32_BCLK
I2S_ESP32_LRCLK
I2S_ESP32_IN_SDATA
I2S_ESP32_OUT_SDATA
ADAU_RESET#
SPI_ADAU_CS
SPI_ADAU_MISO
SPI_ADAU_MOSI
SPI_ADAU_SCLK
I2S_RADIO_IN_BCLK
I2S_RADIO_IN_LRCLK
I2S_RADIO_IN_SDATA
I2S_BT_BCLK
I2S_BT_LRCLK
I2S_BT_IN_SDATA
I2S_BT_OUT_SDATA
I2S_ESP32_BCLK
I2S_ESP32_LRCLK
I2S_ESP32_IN_SDATA
I2S_ESP32_OUT_SDATA
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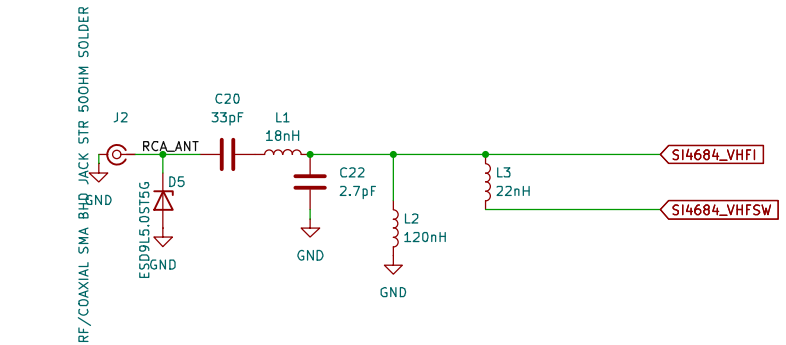

3V3_DIGITAL
RESET1
R1
10K
ESP32_EN
TS3425UA-3x4x2.5-160
C2
1uF
GND
BOOT1
ESP_32_BOOT
C1
100nF
U1
24AA025E64
A0
A1
A2
Vcc
Vss
SDA
SCL
I2C_SDA
I2C_SCL
U3
W25Q128JVS
SPI flash for main filesystem
IO3
IO2
DO(IO1)
DI(IO0)
CLK
CS
VCC
R9
R10
R31
M_SPI_MISO
M_SPI_MOSI
M_SPI_SCLK
M_SPI_CS#
C5
10uF
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SHIELD
SBU1
SBU2
R2
R3
5.1K
D6
SMF6V0A
V_USB
D1
D2
D_TVS
USB_D-
USB_D+
PWR_FLAG
U2
ESP32-S31-WROOM-3
C3
22uF
C4
100nF
3V3
EN
R45
R48
R49
33
SPI_MOSI
SPI_MISO
SPI_SCLK
BT_UART_RX
BT_UART_TX
BT1058_SYS_CTRL
BT1058_RESET
ADAU_RESET
SI4684_RESET
SI4684_INT
SI4684_SMODE
ENABLE_ANALOG_OUT
GPIO61
GPIO60
UART0_RXD/GPIO59
UART0_TXD/GPIO58
MTMS/GPIO57
MTDI/GPIO56
MTCK/GPIO55
MTDO/GPIO54
GPIO53/ADC2_CH1_P
GPIO52/ADC2_CH1_N
GPIO51/ADC2_CH0_P
GPIO50/ADC2_CH0_N
GPIO49/ADC1_CH3_P
GPIO48/ADC1_CH3_N
GPIO47/ADC1_CH2_P
GPIO46/ADC1_CH2_N
GPIO45/ADC1_CH1_P
GPIO44/ADC1_CH1_N
GPIO43/ADC1_CH0_P
GPIO42/ADC1_CH0_N
GPIO40
GPIO39
GPIO38
R55
M_SPI_SCLK
M_SPI_CS#
M_SPI_MISO
M_SPI_MOSI
R7
R8
2.2K
I2C_SDA
I2C_SCL
I2S_IN_ESP32_SDATA
I2S_LRCLK
I2S_BCLK
I2S_OUT_ESP32_SDATA
R17
R19
10K
3V3_AUDIO
R5
SDIO_DATA0/GPIO20
SDIO_DATA1/GPIO21
SDIO_DATA2/GPIO22
SDIO_DATA3/GPIO23
SDIO_CLK/GPIO24
SDIO_CMD/GPIO25
USB_DP
USB_DM
GPIO33
GPIO34
GPIO35
GPIO36*
GPIO37
SPI_CS_ADAUD
SPI_CS_SI4684D
Sheet: /Esp32-S31/
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3V3_AUDIO
DVDD
IOVDD
AVDD
PVDD
10uF
C58
C59
C60
C61
R25
0.47
GND
C47
C48
C62
C63
100nF
C45
C46
C70
C69
C68
C67
C44
C49
C50
C64
C6
1uF
1V2
C65
FB3
BLM18SG121TN1D
J5
FCR684205T
GND
VCC
VIN
SPDIFOUT
R56
33
U13
ADAU1452
PLL_LF
PLLFILT
VDRIVE
SPDIFIN
SPDIFOUT
XTAL_2
XTALIN/MCLK
CLKOUT
XTAL_1
XTALOUT
THD_P
THD_M
I2S_RADIO_IN_BCLK
I2S_RADIO_IN_LRCLK
I2S_RADIO_IN_SDATA
BCLK_IN0
LRCLK_IN0/MP10
SDATA_IN0
LRCLK_OUT0/MP4
BCLK_OUT0
SDATA_OUT0
I2S_BT_LRCLK
I2S_BT_BCLK
I2S_BT_OUT_SDATA
I2S_BT_BCLK
I2S_BT_LRCLK
I2S_BT_IN_SDATA
BCLK_IN1
LRCLK_IN1/MP11
SDATA_IN1
LRCLK_OUT1/MP5
BCLK_OUT1
SDATA_OUT1
I2S_ESP32_LRCLK
I2S_ESP32_BCLK
I2S_ESP32_OUT_SDATA
I2S_ESP32_BCLK
I2S_ESP32_LRCLK
I2S_ESP32_IN_SDATA
BCLK_IN2
LRCLK_IN2/MP12
SDATA_IN2
LRCLK_OUT2/MP8
BCLK_OUT2
SDATA_OUT2
BCLK_IN3
LRCLK_IN3/MP13
SDATA_IN3
LRCLK_OUT3/MP9
BCLK_OUT3
SDATA_OUT3
IOVDD
ADAU_RESET#
SELFBOOT
RESET
SS_M
MOSI_M
SCLK_M
MISO_M
SCL_M/SCLK_M/MP2
SDA_M/MISO_M/MP3
MISO/SDA
SCLK/SCL
MOSI/ADDR1
SS/ADDR0
SPI_ADAU_MISO
SPI_ADAU_SCLK
SPI_ADAU_MOSI
SPI_ADAU_CS
AUXADC0
AUXADC1
AUXADC2
AUXADC3
AUXADC4
AUXADC5
MP6
MP7
PGND
AGND
EP
DGND
U14
25AA1024-I/SM
SS_M
MISO_M
SCLK_M
MOSI_M
CS
SO
HOLD
WP
SCK
SI
VCC
VSS
C32
100nF
4.3k
R46
C81
150pF
C82
5.6nF
R51
100
XTAL_1
XTAL_2
ABM3B-12.288MHZ-10-1-U-T
Y3
C141
C142
12pF
R53
1K
C39
1nF
R54
10K
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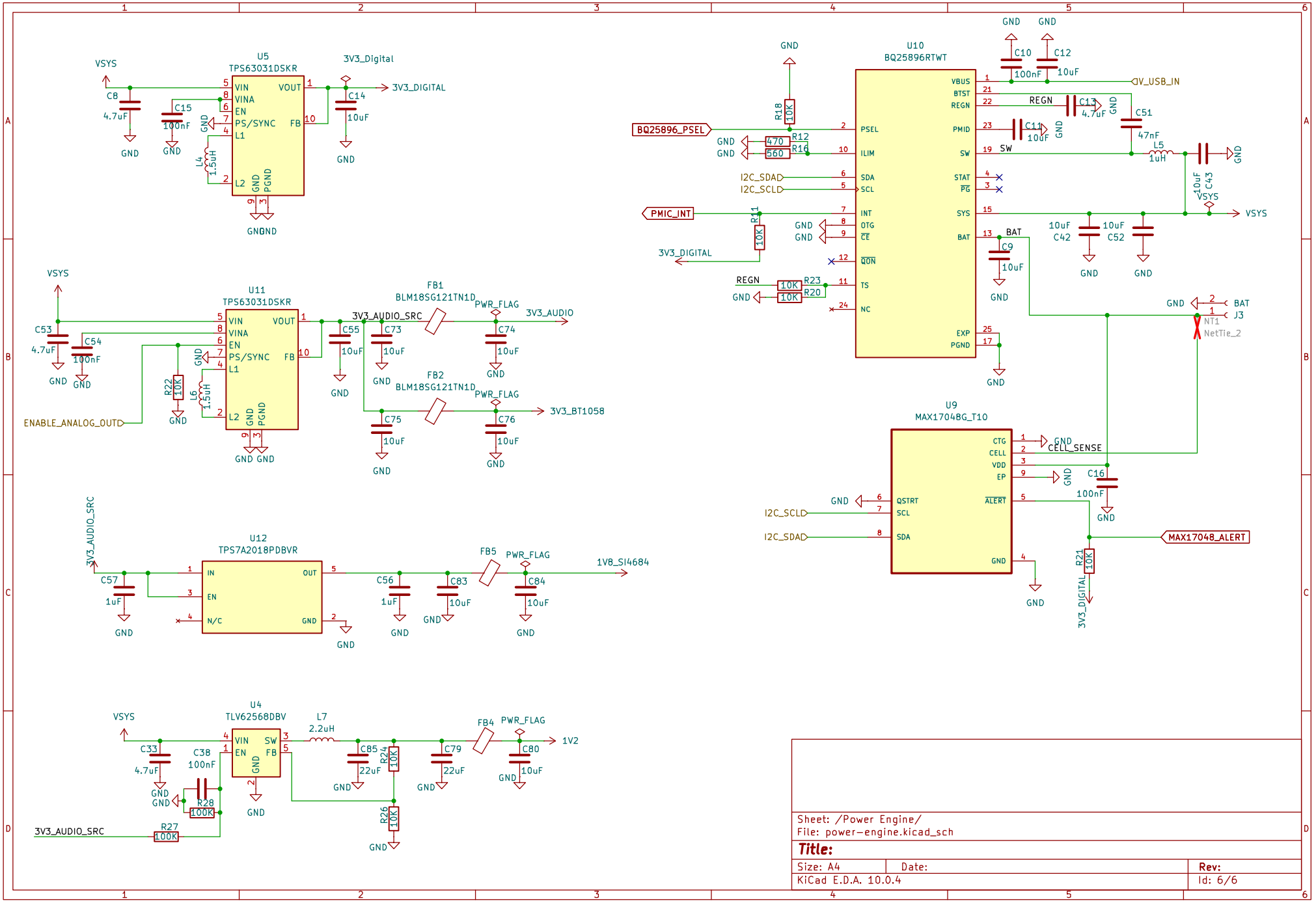


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File: power-engine.kicad_sch

Title:

Size: A4	Date:	Rev:
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